

Fairness and Transparency in ML: A Methodological Framework for Ethical Model Evaluation

Nelson Salazar Valdebenito

NESALAZAR@ALUMNOS.UAI.CL

*Faculty of Engineering & Sciences, Universidad Adolfo Ibáñez,
Santiago 7941169, Chile*

Gonzalo A. Ruz

GONZALO.RUZ@UAI.CL

*Faculty of Engineering & Sciences, Universidad Adolfo Ibáñez,
Santiago 7941169, Chile
Center of Applied Ecology and Sustainability (CAPES),
Santiago, 8331150, Chile
Data Observatory Foundation,
Santiago, 7941169, Chile*

Reinel Tabares-Soto

RTABARES@AUTONOMA.EDU.CO

*Department of Electronics and Industrial Automation,
Universidad Autónoma de Manizales,
Manizales 170001, Colombia
Departamento de Sistemas e Informática, Universidad de Caldas,
Manizales, 170001, Colombia
GobLab School of Government, Universidad Adolfo Ibáñez
Santiago 7941169, Chile
Faculty of Engineering & Sciences, Universidad Adolfo Ibáñez,
Santiago 7941169, Chile*

Abstract

In the times when AI and ML applications are more widely spread across our society, new challenges arise when these are being used by public organisations. Ethical concerns may emerge for various reasons, but specially when these models inherit certain biases that could eventually lead up to hurting and/or punishing people with no reason whatsoever. In this study, a comprehensive methodology is introduced to address ethical considerations in machine learning, particularly focusing on biases and disparities. This paper showcases the application of this methodology through two experiments, each involving two models: a base model without ethical considerations and an improved one. The primary objective was to evaluate and mitigate inequalities and biases in machine learning models, in this case, using data from Chile's Public Criminal Defence Office (DPP), which needed a tool that could help in predicting the outcome of a criminal trial. The results were promising, with improvements observed in both models. Specifically, for one of our experiments, there was a reduction in the false negative rate from 40.54% to 31.63%. Additionally, disparities between attributes were reduced by an average of 44.37%. The methodology not only aids in understanding these ethical challenges but also offers tools to optimize and address them, all while maintaining performance.

1. Introduction

As Machine Learning (ML) and Artificial Intelligence (AI) tools are more common in our society, new challenges arise when these get applied in the public sector – such as in organisations and/or public policies (?). Since these tools are often considered “black boxes”, it is quite difficult to understand exactly how an algorithm arrives at a particular decision.

Ethical issues may arise when considering the inherent biases in the data used to train these algorithms. If the judicial system has historically exhibited bias towards particular demographic groups, an algorithm trained with such data may reinforce or intensify this bias.

The problem is, this can affect people. Going back to the judicial system example, if it has a model that can predict the outcome of a trial, that could influence the final sentence of the defendant, potentially changing their life forever. Of course, not every application of this technology is the same - our social media feed may show us something we do not really want to see, but it would not have a huge impact on our lives.

However, if such technology is implemented in crucial sectors such as healthcare, finance or law enforcement, the ramifications can be considerable and harmful. In these industries, an erroneous forecast or partial verdict could result in an incorrect diagnosis, economic adversity or false imprisonment. It is not exclusively about receiving inaccurate recommendations; rather, it concerns the compromise of fundamental rights, fairness and impartiality.

The use of ML applications in public contexts also raises questions about liability and consent. Who is responsible when an algorithm makes a harmful decision? How can we ensure that these models do not possess risks to our society? How can we be sure that the individuals affected by these decisions were properly informed about the use of such tools?

In fact, one way to address these ethical concerns is through regulation. In June 2023, the European Union approved its first AI regulatory act, which sets out harmonised rules for AI systems. The regulation outlines specific bans on certain AI practices, mandates for high-risk systems, and emphasises transparency in AI-human interactions and media manipulation. It also provides a comprehensive framework for market surveillance, covering both providers and users of AI systems in the EU (?). Governments are not the only ones interested in this, industry is too. OpenAI wants regulation for what it calls “frontier AI” (?), i.e., models whose capabilities could pose serious risks to public safety.

In addition, AI cannot replicate many human capabilities. In the field of law, a lawyer must combine abstract thinking with skills to solve problems in situations of high uncertainty, in both the legal and factual domains; something a classification model cannot yet achieve (?). This is, models cannot “reason” in the same manner as a human being. Even the large language models — such as OpenAI’s ChatGPT or Google’s Bard — that have become so popular recently are not capable of doing so, or at least not independently (?). This is due to the fact that behind that black box, there are only probabilities and computations among vectors and matrices. Hence, models should be “ethically trained” in order to avoid these problems.

It is crucial to recognise that while the potential benefits of AI and ML are enormous, the risks they pose, especially if left unchecked, can have profound implications for individuals and society as a whole.

1.1 Background

These ethical concerns have been previously addressed. Several studies have suggested various guidelines and methodologies for AI ethics (??, ??, ??), but they have not yet converged to a standard in technical and ethical terms (??, ??, ??), as many of these studies are task-specific and focused on AI development rather than on ethical considerations (??, ??).

However, ethical assessment tools can serve several purposes, including identifying disparities and biases in models. These tools, such as What-If-Tool (??), SHAP (??), AI Fairness 360 (??), Fairlearn (??), and Aequitas (??), are effective for analysing model performance, optimising models, and auditing their results. For documentation purposes, there are different tools available such as Model Cards (??) and Datasheets for Datasets (??), among others.

The use of ethical tools and techniques for ML algorithms is not widespread when developing such models. This is largely due to its recent emergence and a lack of obligation for their use by private companies. However, in the public sector, this becomes a more pressing issue.

For instance, a common use case is in models that classify taxpayers, debtors, and financial and tax-related issues. Specifically, Black et al. (2022) (??) conducted a study on income fairness in the tax audit models used by the US’ Internal Revenue Service (IRS). The study examined vertical equity, that is, how much the model represents important individual differences, given their relevance to public tax policies. Using various ethical tools, the study discovered that prioritising return on investment (ROI) from auditing lower-income taxpayers by the IRS compromises vertical fairness in an effort to decrease the expenses associated with such audits.

Healthcare is another crucial field for ML models. Rajkomar et al. (2018) (??) investigated the effects of these algorithms on medical care, particularly how biases and disparities could put more vulnerable groups at risk of incorrect diagnoses or treatments. The research employed distributive justice strategies to guarantee impartial outcomes for patients and ensures that resource allocation is as equitable as possible.

Singh and Joachims (2018) (??) proposed a methodology to ensure fairness in rankings. The study emphasized the prevalence of rankings in various contexts, including books, movies, jobs, and individual comparisons. Therefore, it is essential for classification algorithms to prioritize not only the benefit of the end-user but also the fairness of the ranked entity. The research has presented various probabilistic approaches that act as constraints in diverse situations, like ensuring demographic parity or addressing disparities in data.

Ethical algorithms encompass not only the models’ operations but also their documentation, specifically their transparency. Heger et al. (2022) (??) examined the needs and perspectives of individuals constructing ML models on data documentation, using datasheets for datasets. The study uncovered the absence of standardised methodologies for comprehensive and user-friendly documentation. Furthermore, there remains a disparity in users’ attitudes towards documentation, often seen as an extra burden rather than a tool for facilitating accountable employment of ML models.

Data also have important implications for model training. According to a recent study on the use of ML models to detect COVID-19 in X-ray images from various databases (??), researchers discovered that data imbalances and biases in patient age and sex can lead

to underperformance of the algorithm, which affects the accuracy and generalisability of predictions.

Furthermore, Davidson et al. (2019) (?) determined that numerous Twitter/X datasets exhibit biases towards African American English speakers/writers. This is particularly problematic when applied in technologies for detecting abusive language, as these systems tend to classify these tweets as abusive more frequently than those written in standard English, resulting in discrimination against groups who are often the targets of such language abuse. According to Wiegand et al. (2019) (?), these models’ biases are closely related to the way the data was sampled.

1.2 Key Definitions

Several terms have become crucial when considering the ethical and practical implications of models and their decision-making capabilities. These terms shed light on how models process information, how they can potentially introduce inequalities, and how to interpret their results. Some of these fundamental concepts that are going to be used in this paper are as follows:

1. **Transparency:** This refers to the comprehensibility of an algorithm’s decision-making process. In essence, algorithms should not operate as “black boxes” where decisions are made without clarity. Ideally, we should have a clear understanding, or at least a basic idea, of how the algorithm works.
2. **Bias:** This refers to any systematic bias in the algorithm’s results that favours certain groups over others. For example, an algorithm that consistently predicts that people from a certain demographic group are likely to underperform in a job - regardless of reality - is biased.
3. **Fairness:** This ensures that each group of people is treated fairly by the algorithm. In practical terms, the results of the algorithm should not unfairly discriminate against any particular group.
4. **Disparities:** This term addresses the differences in outcomes or effects that models produce across various demographic groups. Disparities can arise even in the absence of explicit bias, highlighting the nuanced nature of fairness. It underscores the importance of evaluating the broader impacts of algorithms across different sections of the population to ensure equitable results.
5. **SHAP Value:** Comes from SHapley Additive exPlanations. Is a game-theoretic approach to explaining the outputs of ML models. Inspired by Shapley values, they assign the contribution of each feature to a prediction, ensuring a fair distribution. This provides a better understanding of model interpretability and individual predictions (?).

1.3 Research Context & Paper Structure

This paper was produced as part of the Ethical Algorithms (*Algoritmos Éticos*) project, led by the GobLab of the University Adolfo Ibáñez and developed in collaboration with

the Inter-American Development Bank (IDB), which aims to incorporate ethical standards into the development of tools based on ML and AI models. These ethical standards aim to improve the levels of transparency, fairness, privacy, explainability, and accountability of ML models. It also seeks to establish new purchasing standards for these tools in public tenders.

This project focuses primarily on those performed by or for public institutions and/or state agencies. In particular, this paper is part of the work carried out for the Public Defence Office (DPP) of Chile, a public body that provides defence services to anyone who does not have a lawyer to defend them in a criminal trial. The DPP needed a predictive system to help predict the outcome of a trial in order to carry out audit and quality control tasks on its defence lawyers.

The results shown in this paper correspond to the application of a framework developed for this purpose, which is used to evaluate inequalities and biases in ML models. This framework aims not only to help analyse and understand these problems, but also to mitigate them by using different tools to optimise and address them. The methodology was put into test by doing two experiments, each one with two models: a base model with no ethical considerations, and an optimised one. In the end, both models were improved in both disparities and biases, whilst having a minimal impact on performance. Our best results include a false negative rate reduction 40.54% to 31.63%, and its disparities between attribute reduced by an average of 44.37% for one of the experiments.

This paper is structured as follows. Section 2 presents the proposed methodology for this project, as well as the datasets and tools that were used. In section 3, the experiments, models and results are shown; whilst in section 4 those results are discussed. Finally, section 5 contains the conclusion and future work.

2. Methodology

We propose a comprehensive framework for evaluating and enhancing ML models in terms of fairness and bias, applicable across different techniques like gradient boosting machines and neural networks. This framework is designed to answer key questions on model performance, improvements, and functioning in the context of fairness and bias.

It employs a two-part experiment format, contrasting a base model with an ethically improved model that addresses fairness and biases. The framework emphasises pre- and post-training phases, and encompasses elements of analysis and documentation for a complete ethical model evaluation.

The proposed framework follows a five-step process:

1. Perform basic Exploratory Data Analysis (EDA) to identify data characteristics, potential disparities, and class imbalances.
2. Evaluate the base model regarding its performance, biases (through counterfactual analysis), and fairness (disparities).
3. Using the information obtained, develop an improved model factoring in fairness and ethical considerations, including mitigating potential disparities, balancing sample and class weights, optimising hyperparameters, and selecting the optimal classification threshold.

4. Evaluate the newly created model and compare it to the base version.
5. Apply documentation frameworks and other tools to enhance transparency and understanding of the model’s workings and decision-making process.

2.1 Problem description

As it was previously mentioned, the problem comes from Chile’s Public Criminal Defense Office (*Defensoría Penal Pública*, DPP). This entity needed a tool based on ML techniques that could help in predicting the outcome of a trial, so that they could conduct a series of audits and quality control tests on its lawyers.

Since this implies the use of sensitive data from people processed by different crimes, it requires some kind of ethical analysis of the models developed for this purpose, specially since they are trying to predict an outcome that could potentially change the life of the defendant. In particular, every model developed has one class to predict: if a person has a favourable (1) or unfavourable (0) outcome to its trial. An unfavourable outcome occurs when the defendant is imprisoned and/or its sentence gets increased. A favourable outcome is anything but the previous mentioned – i.e.: sentence reduced, paroled or acquitted.

One of the main concerns that can arise is the way in which the model(s) handles the predictions. This is because of the potential biases that may arise due to the nature of the crime - i.e., its prevalence in certain areas of the country, or by whom it is committed. In that regard, it is important to determine whether one or more models (one per crime) should be developed. A single model would facilitate the development process, but because the crimes may be so different, it may have problems of accuracy, which would mean that it would not be very effective for its purpose. On the other hand, with multiple models, this shall not be a problem, as each would be a specialist in one crime. This can also facilitate ethical analysis, as it allows for the model to inherit any biases and differences that might be found within a given crime.

Similarly, another concern that may arise is the way in which these models are used. While they are intended to be used for auditing purposes, it cannot be ruled out that a lawyer might use them to determine whether or not it is in his or her interest to defend an accused person. It is important to have clear documentation that correctly defines the purpose of these tools, how they work and a clear definition of the potential risks associated with their use.

2.2 Databases

We are using two of the datasets that were handed over for this work. These contains the records of multiple individuals accused of drug trafficking or petty theft in between the years 2017 and 2022. The details are as follows:

- There are 16,690 non-null rows for the drug trafficking set; whilst the petty theft has 39,954 non-null rows.
- Each dataset has a mixture of defendant-related and trial-related columns. The second ones are directly related to the outcome of the trial, so those cannot be used for training purposes. Hence, we are using those related to the defendant: region

(geographical location; “Region”), level of development of the crime (“Desarrollo”), hearings (“Audiencias Efectivas”) and the Barrister (“Defensor”); and exclusively for the drug trafficking set, there is also the foreigner (“Extranjero”) column.

- However, there are many critical variables (such as the age, sex, and previous cases of the defendant, among others) that were not available in these sets. The absence of these features obviously affects the performance that a model can have, but it also limits the capabilities of analysing potential biases and disparities.
- Because of this, a new “Edad” (age) is created for each set. For the drug trafficking one, we use the information from the Fourteenth National Drug Survey in the General Population of Chile, made by the National Service for the Prevention and Rehabilitation of Drug and Alcohol Consumption (?) in Chile, which states that people between the ages of 18 and 34 years old concentrates the majority of the drug consumption in the country. Hence, for this experiment, we take a normal distribution with a mean of 26 (the average between 18 and 34) and a standard deviation of 8.
- For the second dataset, we use the data from the Centre for Crime Studies and Analysis (?) for the crime of theft between the years 2017 and 2022. This variable also considers a normal distribution with a mean of 31 (average age between 18 and 44, which accounts for the majority of those charged with this crime) and a standard deviation of 8.
- There are no visible class imbalances in the first set. However, in the second one there are some important differences. In this case, there are 28,819 unfavourable outcomes, versus 11,135 favourable outcomes. This means that there are 2.5 times more unfavourable outcomes than favourable ones.

2.3 Development & ethical tools

In this section, we look at the essential development and ethical tools used throughout our research. Those in the first category are as follows:

1. **Python:** A versatile and widely-used programming language, Python offers extensive libraries and frameworks for ML and data analysis, facilitating the design and implementation of our algorithms (?).
2. **Data processing and visualisation:** This includes libraries such as Pandas (?) and Numpy (?) for data manipulation, analysis, and transformation, providing a foundation for our data preprocessing needs. For visualisations, we use Matplotlib (?) for basic graphs and plots.

Optimisation tools:

1. **Scikit Learn:** An open-source ML library for Python, Scikit Learn provides simple tools for data analysis and modelling, making it easier to implement ML algorithms (?).

2. **AI Fairness 360**: An extensible open-source toolkit that offers a comprehensive set of fairness metrics for datasets and ML models, as well as algorithms to mitigate biases.
3. **Fairlearn**: A tool that focuses on assessing and improving fairness in ML models, offering mitigation algorithms and fairness metrics.

Analytical and documentation tools:

1. **What-If-Tool**: An interactive visual interface that provides in-depth exploration capabilities for ML models, allowing users to understand their predictions more profoundly.
2. **SHAP**: Offers a game-theoretic approach to model explanations, providing clarity on feature contributions to individual predictions, enabling more transparent model interpretability.
3. **Aequitas**: An open-source bias audit toolkit, Aequitas facilitates the process of understanding and interpreting disparities in ML models across various groups.
4. **Model Cards**: A framework designed to provide comprehensive and standardised reporting on ML model performances, ensuring transparency in their capabilities and limitations across diverse operational environments and user groups.

3. Experiments & Results

3.1 Experiments

Two experiments were conducted, one with the drug trafficking set, and the second one with the petty theft set. Each experiment was divided into two parts: the base situation and the improved situation.

As it was previously mentioned, the purpose of these models is to predict if a defendant will have a favourable or unfavourable outcome to its trial. A set can have one or several outcomes (i.e. “Outcome 1”, “Outcome 2”, etc), which can occur in different instances. For the sake of simplicity, the models developed in here will always take the last outcome available. For the drug trafficking set, it will be the second outcome; for the petty theft dataset, it will be the first (and only) one.

1. The base situation consists of a basic model that can predict the previous mentioned classes. Once the model is created, the results are analysed using SHAP values (?), the What If Tool (?) and Aequitas (?), so that the biases and disparities get identified.
2. In the improved situation, as its name suggests, the model gets refined upon the insights gathered in the first part. This also considers the selection of a protected feature. In this case, for both experiments, it is going to be the “Region” feature.
 - (a) For the pre-training phase, we use AI Fairness 360 (?) and its Reweighting tool to assign new weights to the sample data, as well as Scikit-Learn’s (?) *compute_class_weight* function for optimising the class weights. Some basic hyperparameter tuning was also applied for some gains in model performance.

- (b) For the post-training phase, we use Fairlearn’s *ThresholdOptimizer* function for optimising the prediction threshold of our model. The constraint used for this is the false negative rate parity, since we are interested in minimising the amount of people that has a non-favourable ending to its trial. After that, the model gets analysed using the same tools mentioned before. Additionally, it can be documented using Google’s Model Cards framework for model documentation.

For the protected variable, we want to select those regions that are under-represented in the dataset. There are a total of 16 regions in Chile, and since we want to ensure a fairly equal representation of these places, in both experiments the regions will be sorted by number of records (highest to lowest), and half of them will be selected (8) as favored and/or unfavored.

Figure ?? shows a detailed flowchart for the proposed framework that is applied for these experiments:

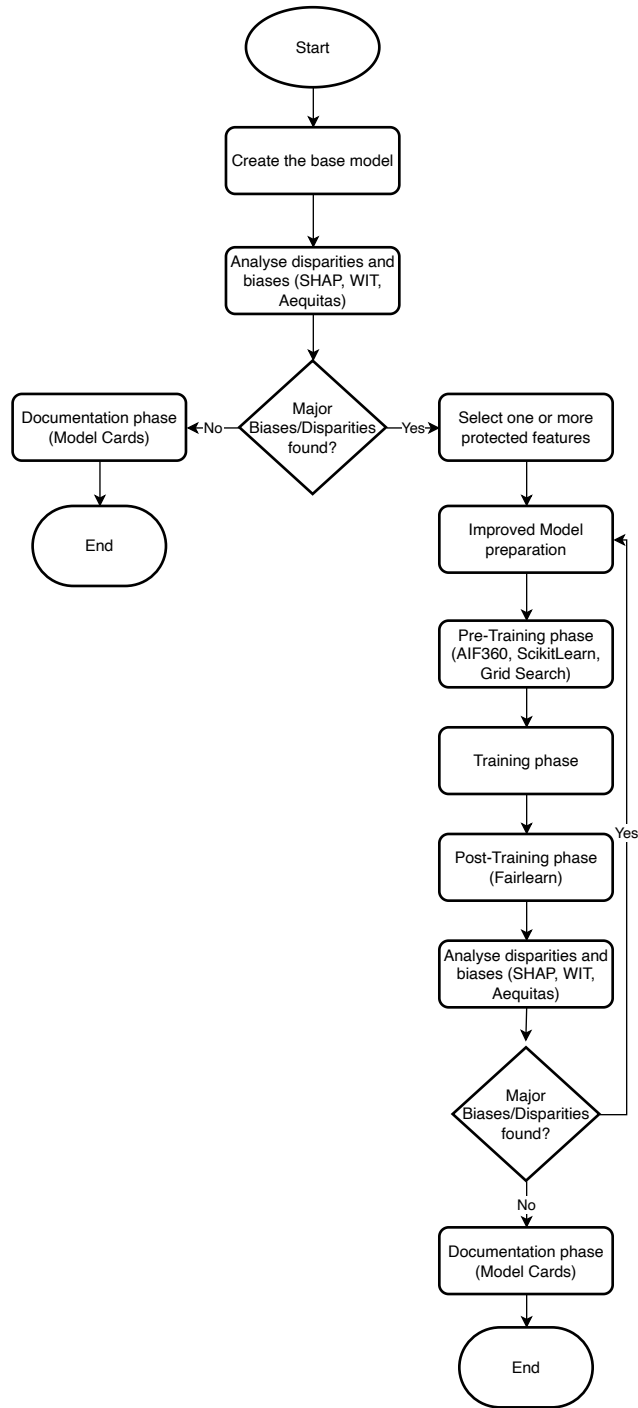


Figure 1: Proposed framework flowchart.

3.2 Results: Drug trafficking

For this experiment, a multi-layer perceptron was used. The details of the architecture and training for the base situation are as follows:

- **Architecture:** Multi-Layer Perceptron (MLP), developed with Tensorflow, and trained to minimise the binary crossentropy.
- **Type:** Classification. In this case, to predict if a person has a favourable ending (1) or not (0).
- **Datasets:** 70% split for the training set, 30% for the testing set, and a 1000 rows were removed from the training dataset to create a validation dataset. In total, there are 10683 rows for training, and 5007 for testing.
- **Hyperparameters:**
 - 70 training epochs.
 - Learning rate: 0.01
 - Batch size: 1024.
 - Layers: 4 - One input layer, two hidden layers and one output layer.
 - Neuron configuration: 6, 13, 5, 1.
 - Activation functions: Sigmoid.

For the second part of this experiment, the improved situation, there is a similar configuration with a few changes, those being the amount of neurons for the hidden layers: 14 for the first one, and 12 for the second one, just for testing purposes; as well as both hidden layers now using a ReLU activation function.

This new model also uses the improved sample weights that were calculated by AIF360’s reweighting function. The class weight were also balanced using Scikit-Learn’s *compute_class_weight* method, even though it was not really necessary to do so. Model performance for both models are shown in Table ??.

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	80%	100%	62%	77%
Base Test	78%	100%	59%	75%
Improved Train	67%	67%	72%	70%
Improved Test	65%	67%	68%	68%

Table 1: Model performance.

3.2.1 SHAP VALUES

SHAP values were calculated by using the data from the training set for both models. Table ?? presents the absolute SHAP values, indicating the overall effect on the model, whereas Table ?? provides information about its impact on the predictions (if it contributes towards positive or negative predictions).

Model	Barrister	Age	Region	Development	Hearings	Foreigner
Base	0.006392	0.001494	0.016019	0.000132	0.000645	0.330375
Improved	0.051984	0.004848	0.046972	0.000172	0.002176	0.217328

Table 2: Average absolute SHAP Values by feature.

Model	Barrister	Age	Region	Development	Hearings	Foreigner
Base	-0.000258	0.000114	0.000420	0.000132	0.000057	-0.027403
Improved	0.003826	-0.001084	0.009636	0.000172	0.000040	0.017798

Table 3: Average SHAP Values by feature.

Figure ?? and Figure ?? should be interpreted in the following way. The X-axis represents the real SHAP value and its effect on predictions. Negative SHAP values, on the left side of zero, indicate that the variable contributes to a negative prediction by the model. Positive SHAP values, on the right side of zero, indicate that the variable contributes to positive predictions. The feature value is indicated by the colour, where blue denotes lower values and red higher ones.

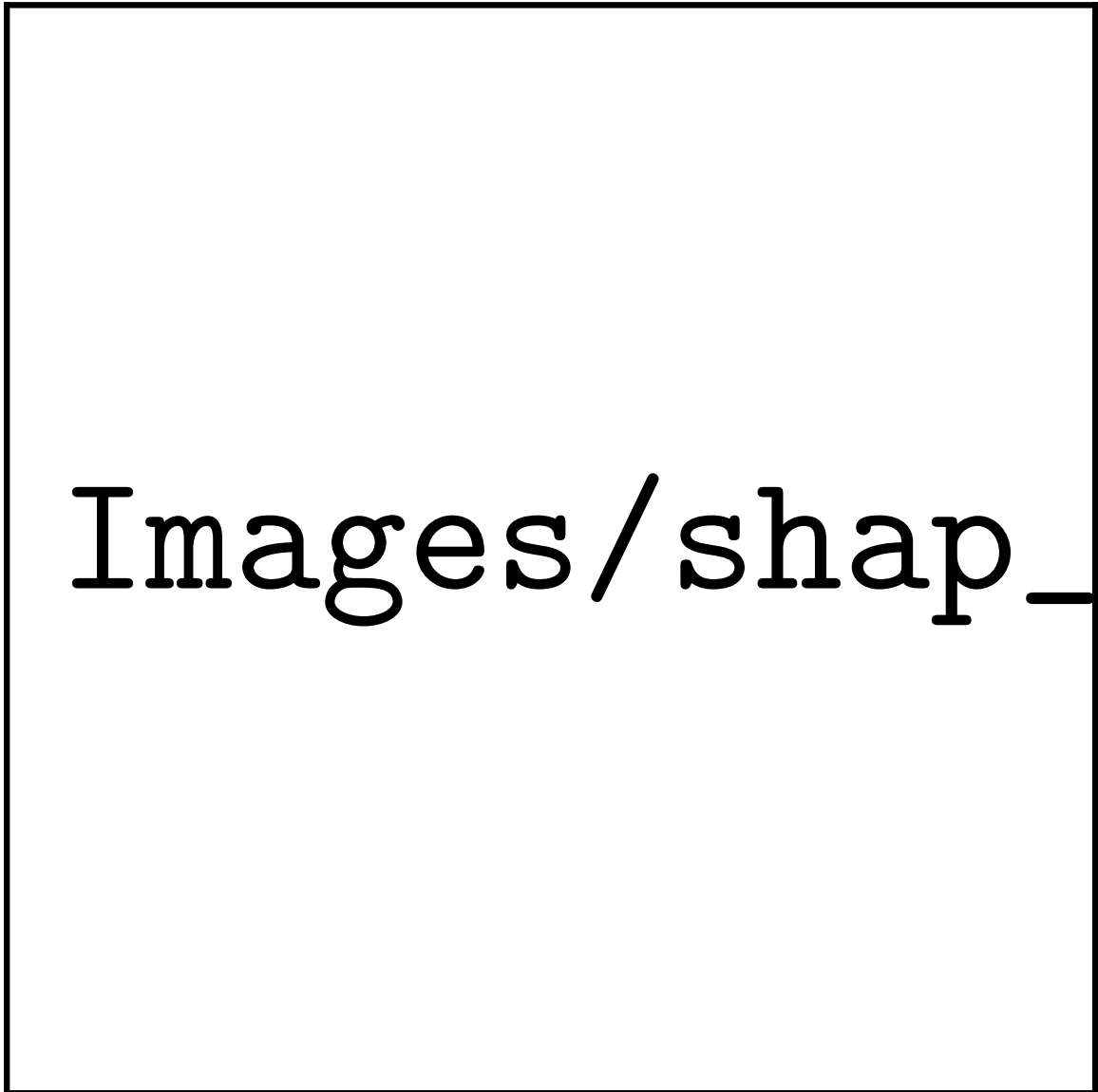


Figure 2: SHAP values distribution by feature (base model).

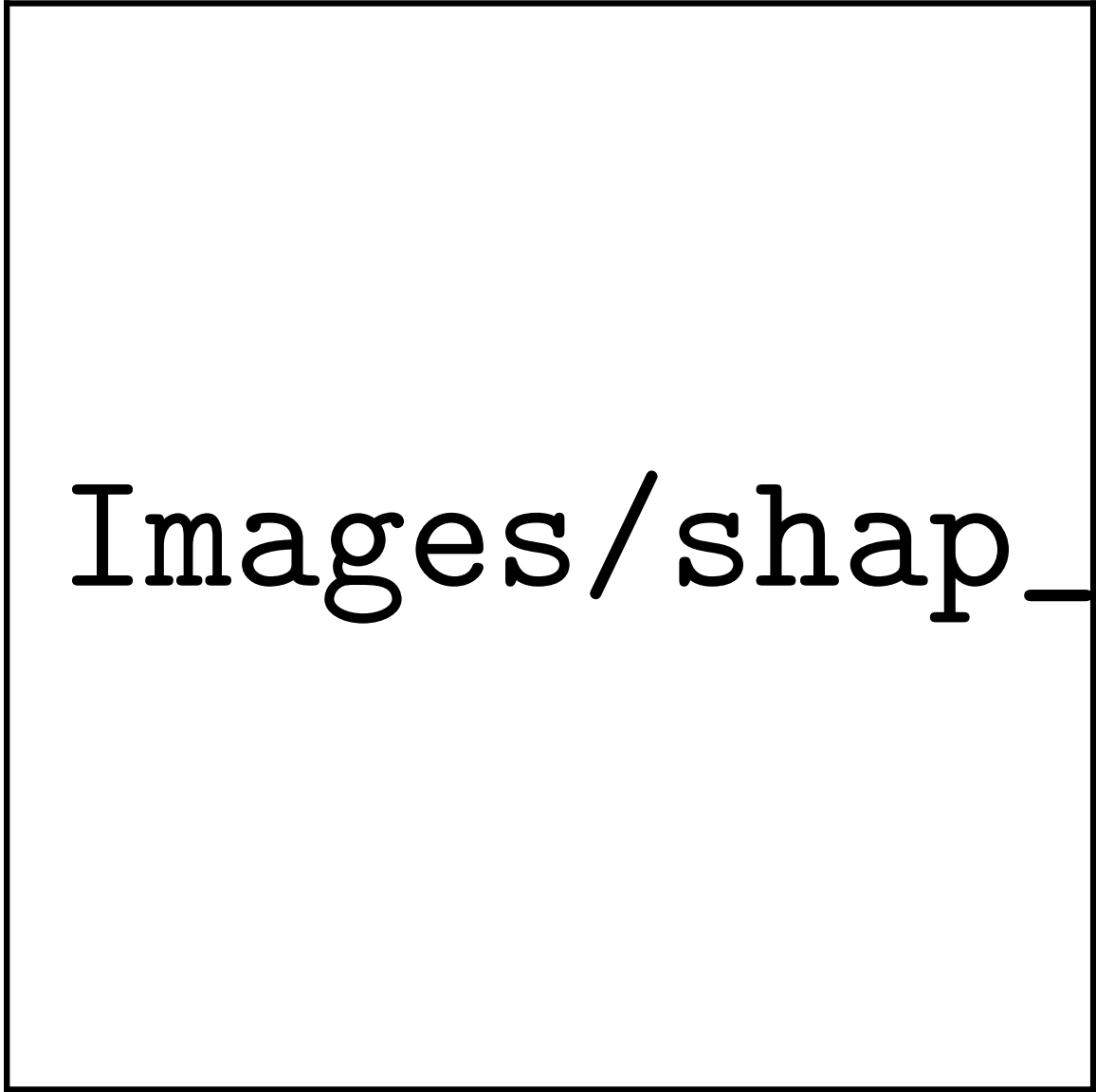


Figure 3: SHAP values distribution by feature (improved model).

3.2.2 FAIRNESS & DISPARITIES (AEQUITAS)

For this part of the experiment, a new column was added for analysis purposes. This is “Zona”, or the zone of the country, as Chile can be divided into three different zones according to its region: north, central and south. This can be useful for analysing the impact of the model in a broader way than “Region”.

The overall metrics for the False Positive Rate (FPR), False Negative Rate (FNR), False Discovery Rate (FDR) and False Omission Rate (FOR) between attributes are shown in Table ???. The confusion matrix for both models can be found on Table ??.

Model	FPR	FNR	FDR	FOR
Base	0.13%	40.54%	0.19%	31.88%
Improved	39.25%	31.63%	33.25%	37.51%

Table 4: Bias metrics between attributes.

Model	True Positives	True Negatives	False Positives	False Negatives
Base	1594	2323	3	1087
Improved	1833	1413	913	848

Table 5: Confusion Matrix.

In terms of disparities, three assessments were made using the different reference groups that Aequitas admits. These are the following: the major group for each feature, the group with the lowest metrics, and a predefined group. In this case, it was set to a defendant from the Metropolitan Region (central zone), aged between 18 and 28 years old, Chilean (not foreigner), whose felony was consummated.

These assessments were made considering the True Positive Rate, the False Omission Rate and the False Negative Rate. The TPR is of interest to us to see how positive predictions are distributed within the groups of each feature; FOR is useful to see where the model is failing to make correct predictions; while the FNR aids in understanding the proportion of false negatives over the total actual positives. Finally, our disparity tolerance (τ) is set to 1.5 times larger or smaller than the size of each reference group.

Table ??, as well as Figures ??, ?? and ?? are the assessment results for the base model; while Table ?? is the crosstab for the variable “Region”, broken down by attribute value. Columns “TPR”, “FNR”, “FOR” and “FDR” are the previously mentioned bias metrics.

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	0.81	1.04	21.005
Major Group	7.53	1.61	19.96
Min. Metrics Group	9.15	2.04	28.12

Table 6: Average disparities for the base model.

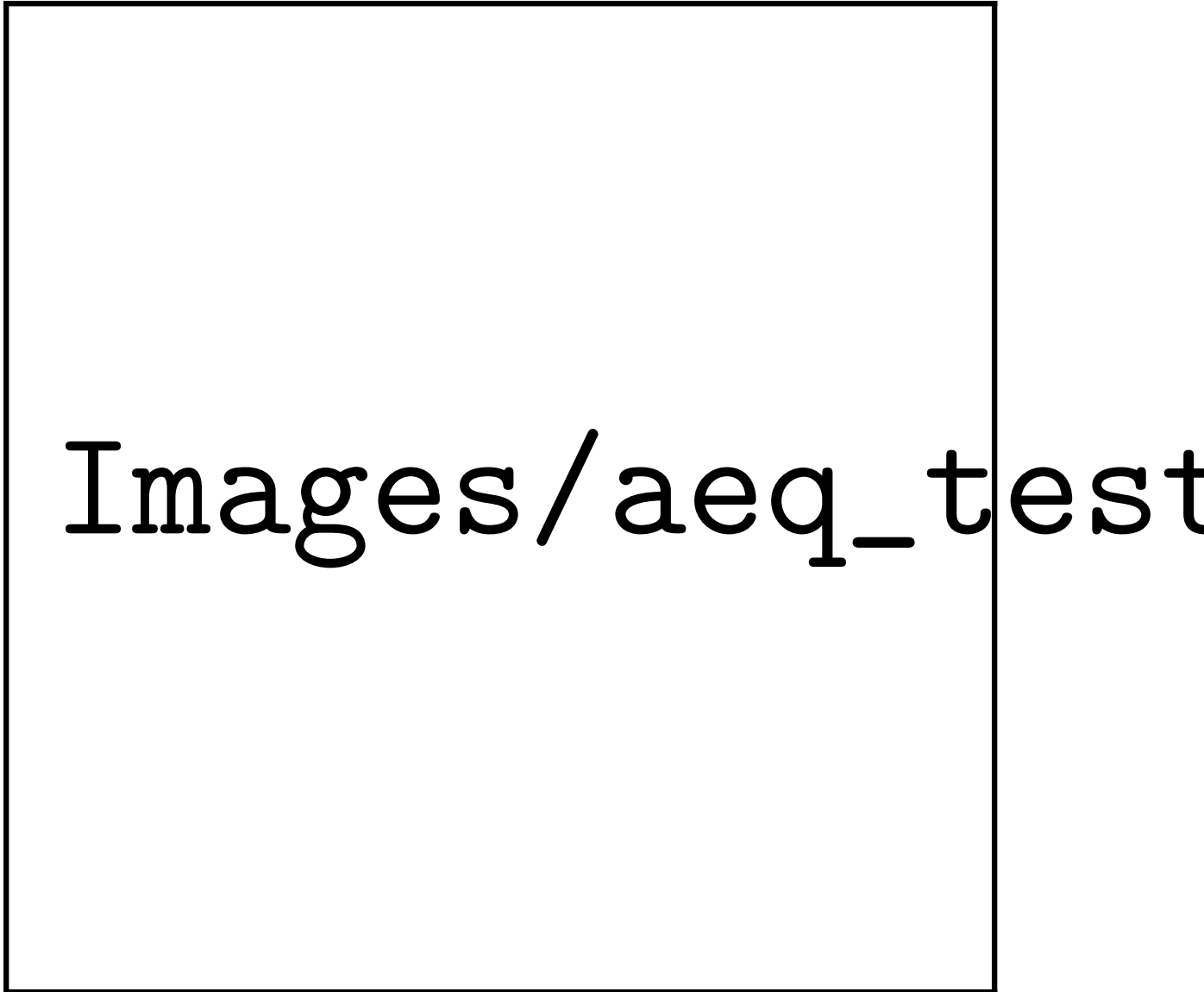


Figure 4: Model assessment for the reference group.

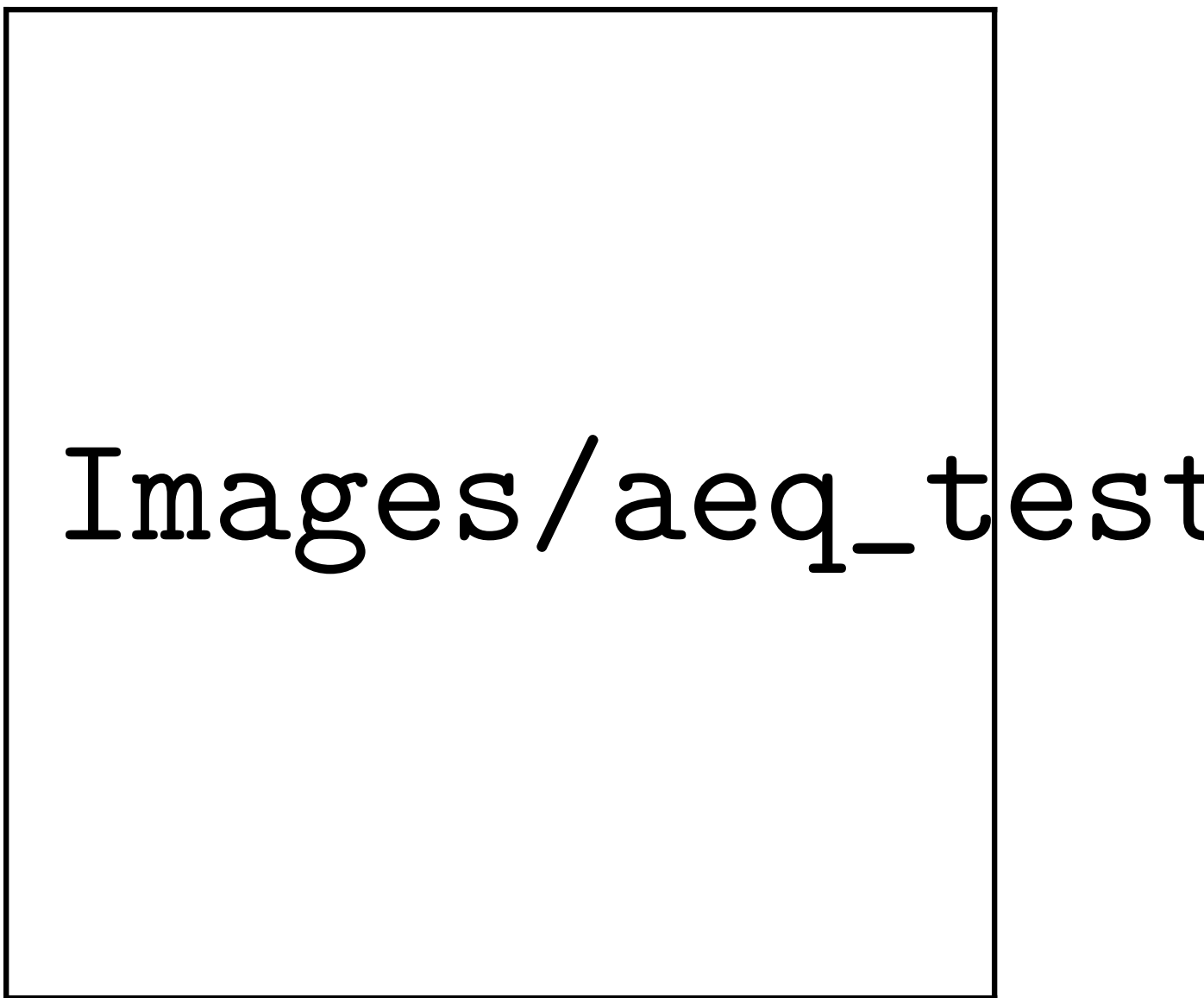


Figure 5: Model assessment for the major group.

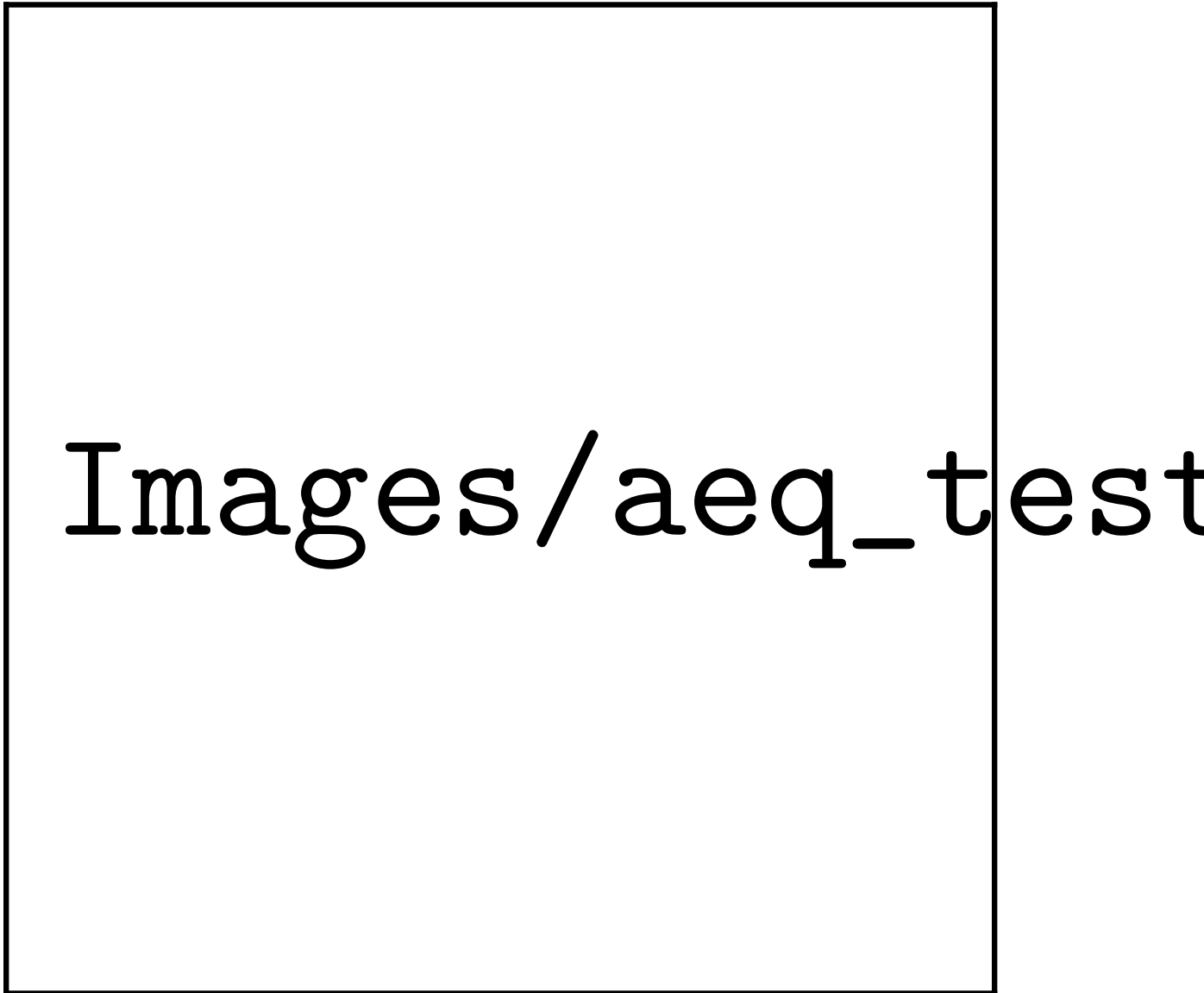


Figure 6: Model assessment for the min metrics group.

Table ??, along with Figures ??, ?? and ?? are the assessment results for the improved model. Table ?? is the crosstab for the variable “Region”.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.8213	0.1787	0.2167	0.0000
Arica y Parinacota	0.7276	0.2724	0.2562	0.0000
Atacama	0.6782	0.3218	0.2456	0.0000
Aysén	0.0000	1.0000	0.3889	-
Bio Bío	0.0139	0.9861	0.4057	0.0000
Coquimbo	0.2547	0.7453	0.5852	0.0000
La Araucanía	0.0556	0.9444	0.3238	0.0000
Libertador Bernardo O'Higgins	0.0566	0.9434	0.3759	0.2500
Los Lagos	0.0227	0.9773	0.3554	0.0000
Los Ríos	0.0000	1.0000	0.1818	-
Magallanes y Antártica Chilena	0.0909	0.9091	0.3448	0.0000
Maule	0.0156	0.9844	0.5000	0.0000
Metropolitana	0.1525	0.8475	0.3067	0.0357
Tarapacá	0.9377	0.0623	0.1720	0.0000
Valparaíso	0.1117	0.8883	0.3911	0.0000
Ñuble	0.0833	0.9167	0.4000	0.0000

Table 7: Average bias metrics between attributes for the base model.

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.05	1.21	0.99
Major Group	1.06	0.87	0.99
Min. Metrics Group	2.12	2.20	1.78

Table 8: Average disparities for the improved model.

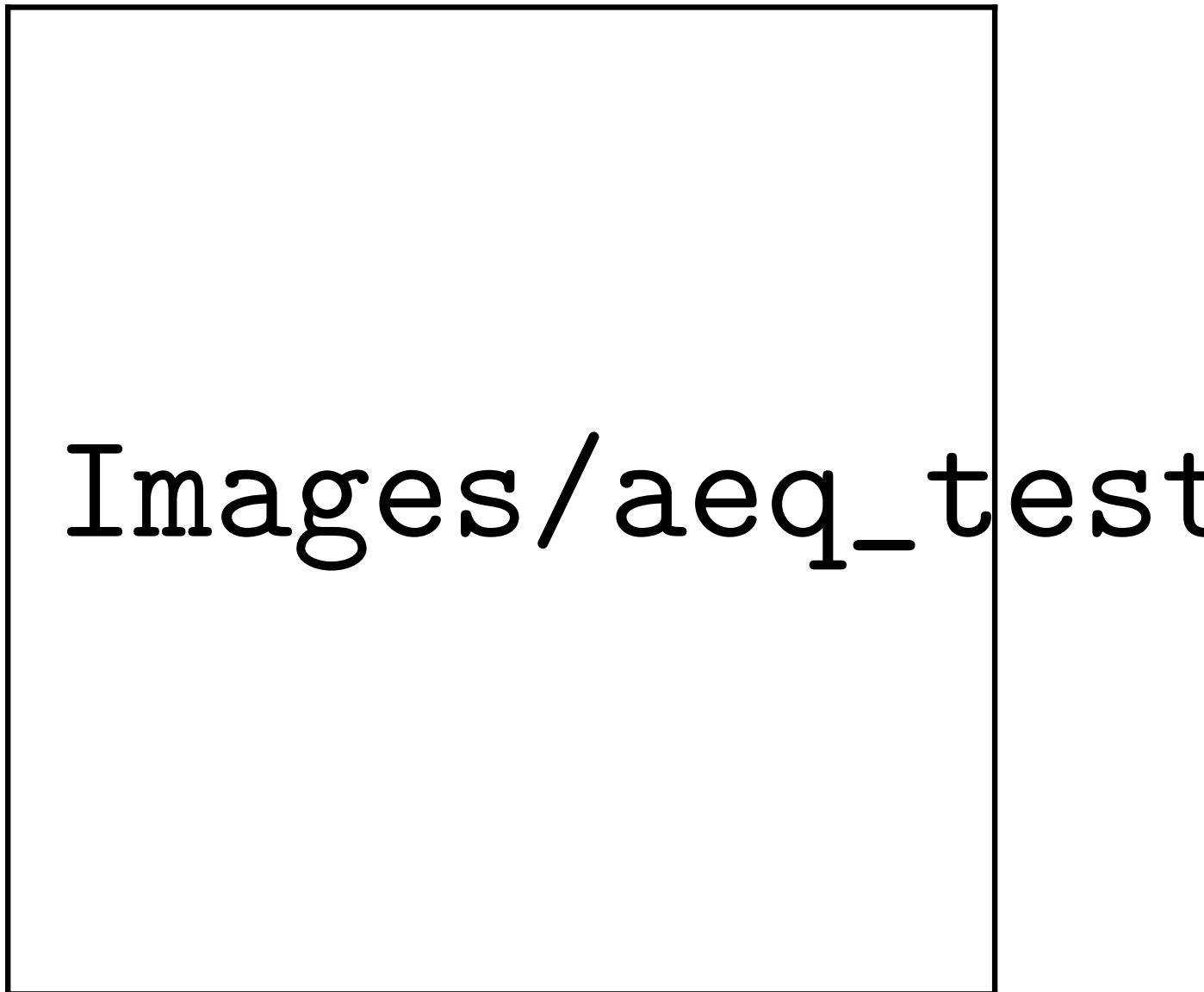


Figure 7: Model assessment for the reference group.

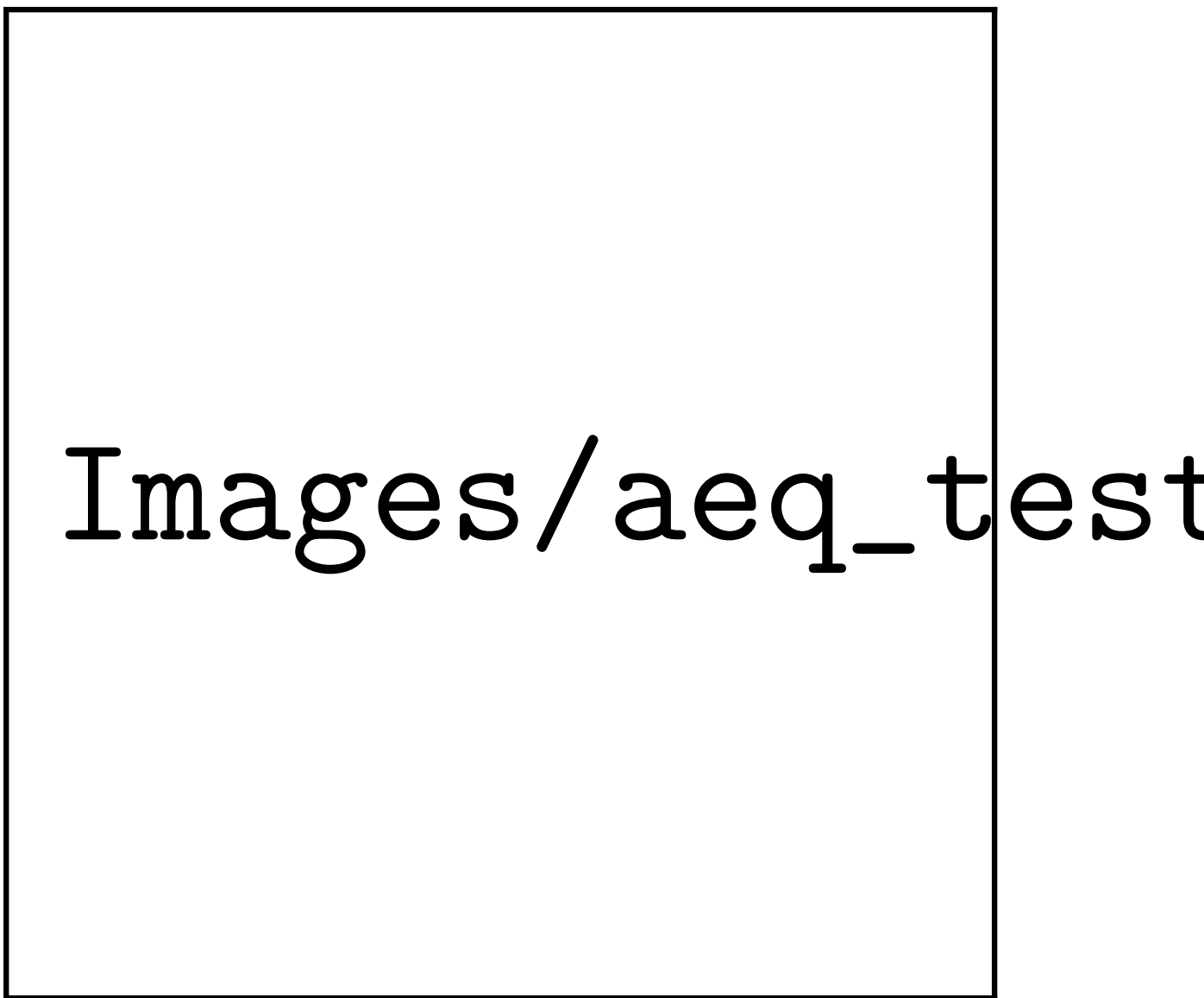


Figure 8: Model assessment for the major group.

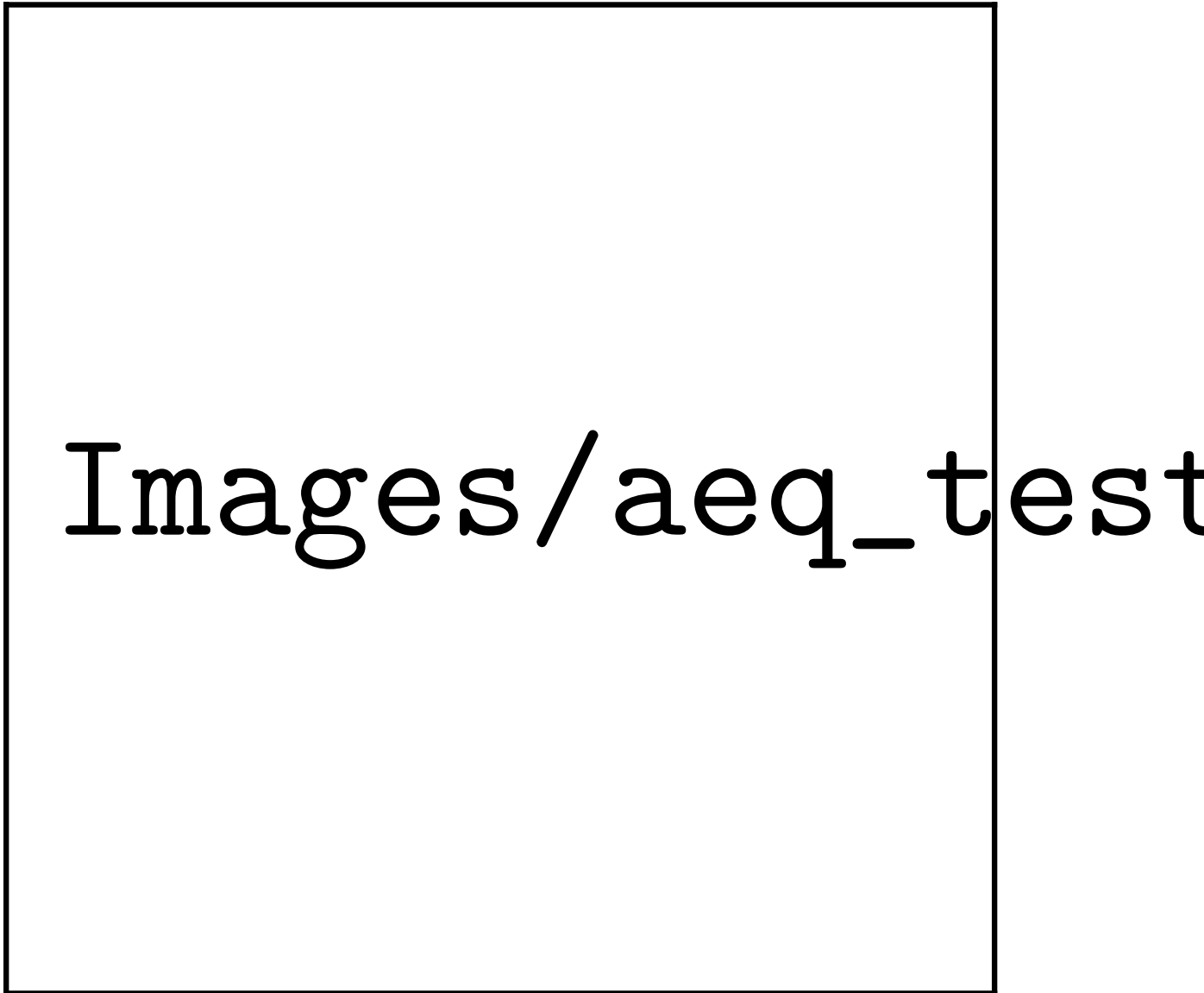


Figure 9: Model assessment for the min metrics group.

3.3 Results: Petty Theft

The second experiment has a different model architecture, in this case using a gradient-boosting framework. The details for the base situation are as follows:

- **Architecture:** Light Gradient-Boosting Machine (LGBM), using the *lightgbm* library.
- **Type:** Classification. In this case, to predict if a person has a favourable ending (1) or not (0).

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.7113	0.2887	0.3088	0.0000
Arica y Parinacota	0.7276	0.2724	0.2562	0.0000
Atacama	0.7126	0.2874	0.2427	0.1143
Aysén	0.3571	0.6429	0.4091	0.6429
Bio Bío	0.7083	0.2917	0.4286	0.5984
Coquimbo	0.6226	0.3774	0.6154	0.3196
La Araucanía	0.4167	0.5833	0.4286	0.7414
Libertador Bernardo O'Higgins	0.7925	0.2075	0.2895	0.5758
Los Lagos	0.5682	0.4318	0.3800	0.6528
Los Ríos	0.5000	0.5000	0.1818	0.8182
Magallanes y Antártica Chilena	0.2727	0.7273	0.3333	0.5000
Maule	0.5938	0.4062	0.5098	0.5000
Metropolitana	0.6949	0.3051	0.2935	0.6306
Tarapacá	0.6955	0.3045	0.5040	0.0000
Valparaíso	0.6383	0.3617	0.3598	0.5367
Ñuble	0.6250	0.3750	0.5000	0.6150

Table 9: Average bias metrics between attributes for the improved model.

- **Datasets:** 70% split for the training set, 30% for the testing set. In total, there are 28967 rows for training, and 11987 for testing.
- **Hyperparameters:**
 - Boosting type: gbdt
 - 100 estimators.
 - 63 leaves per estimator (tree).
 - Unlimited maximum depth.
 - Learning rate: 0.01

For the improved model, the amount of leaves per tree goes up to 120 after doing a basic grid search optimisation. It also considers the new sample weights calculated by AIF360, as well as the new class weights from the *compute_class_weight* function. Model performance is shown in Table ??.

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	78%	77%	33%	46%
Base Test	76%	67%	27%	38%
Improved Train	77%	62%	43%	51%
Improved Test	74%	53%	35%	42%

Table 10: Model performance.

3.3.1 SHAP VALUES

SHAP Values were calculated using the data from the training set for both models. The averages are laid out in Table ?? and Table ??.

Model	Barrister	Age	Region	Development	Hearings
Base	0.060050	0.004756	0.186943	0.167577	0.377383
Improved	0.088094	0.009539	0.183160	0.188467	0.374409

Table 11: Average absolute SHAP Values by feature.

Model	Barrister	Age	Region	Development	Hearings
Base	-0.019104	0.000492	0.023519	0.010751	-0.027672
Improved	-0.016245	-0.000893	0.010753	0.012887	-0.031332

Table 12: Average SHAP Values by feature.

The distribution of these values can be found in Figures ?? and ??.

3.3.2 FAIRNESS & DISPARITIES (AEQUITAS)

The same variable – Zone – was added to this experiment. The overall metrics for the False Positive Rate (FPR), False Negative Rate (FNR), False Discovery Rate (FDR) and False Omission Rate (FOR) are shown in Table ?. The confusion matrix for both models can be found on Table ?.

Model	FPR	FNR	FDR	FOR
Base	4.96%	73.20%	32.98%	22.46%
Improved	12.21%	64.16%	47.54%	21.56%

Table 13: Bias metrics between attributes.

Model	True Positives	True Negatives	False Positives	False Negatives
Base	878	8279	432	2398
Improved	1174	7647	1064	2102

Table 14: Confusion Matrix.

The assessments made for this part of the experiment were the same – one with the major group for each feature, the group with the lower measures and a reference group. The metrics used for this task were also the same: FNR, FOR and TPR disparities; and the results for the base model can be found in Table ? and in Figures ?, ? and ?. Table ? is the crosstab for the variable “Region”.

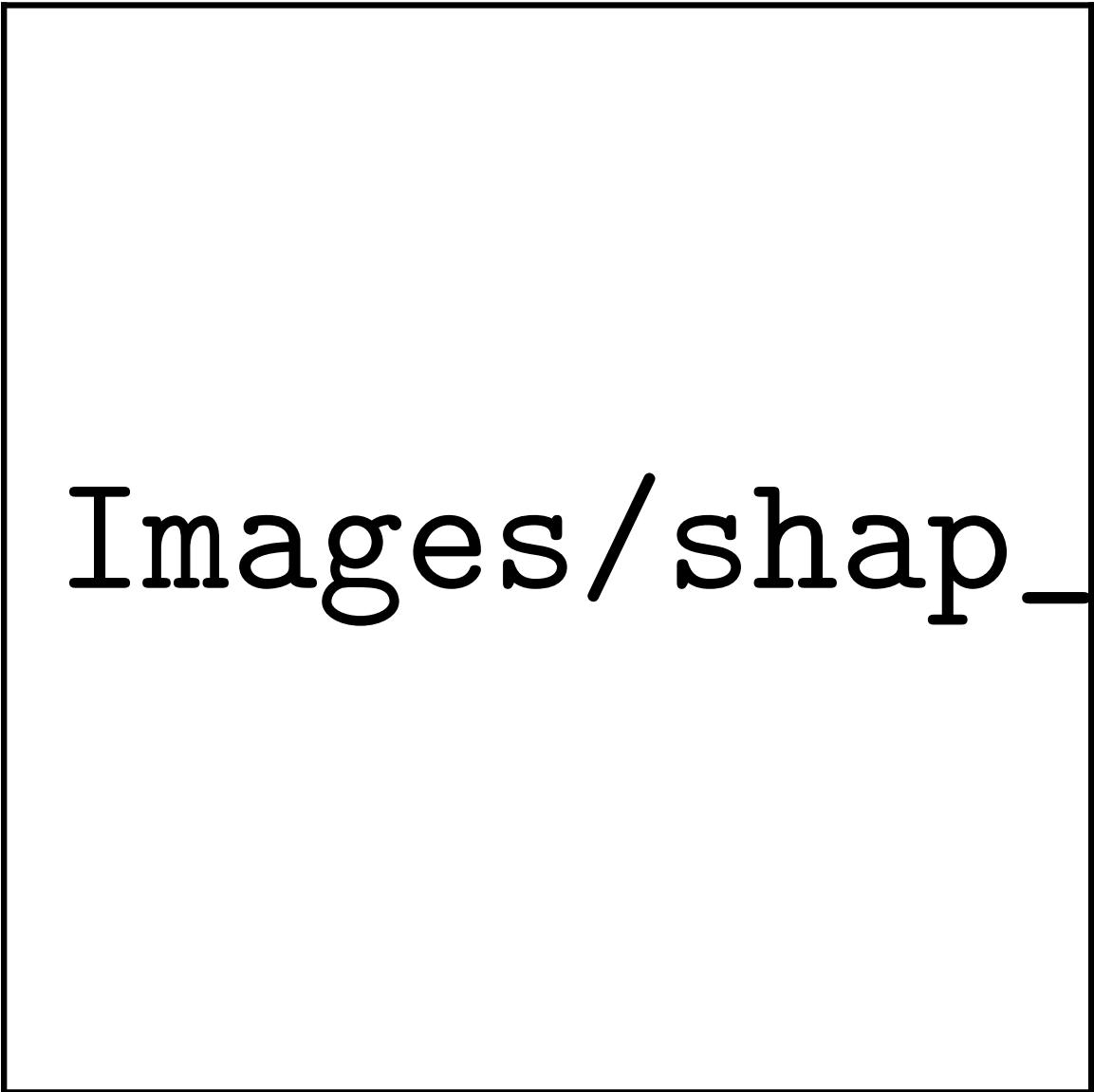


Figure 10: SHAP values distribution by feature (base model).

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.273	1.165	0.644
Major Group	1.273	1.165	0.644
Min. Metrics Group	2.170	2.899	6.272

Table 15: Average disparities for the base model.

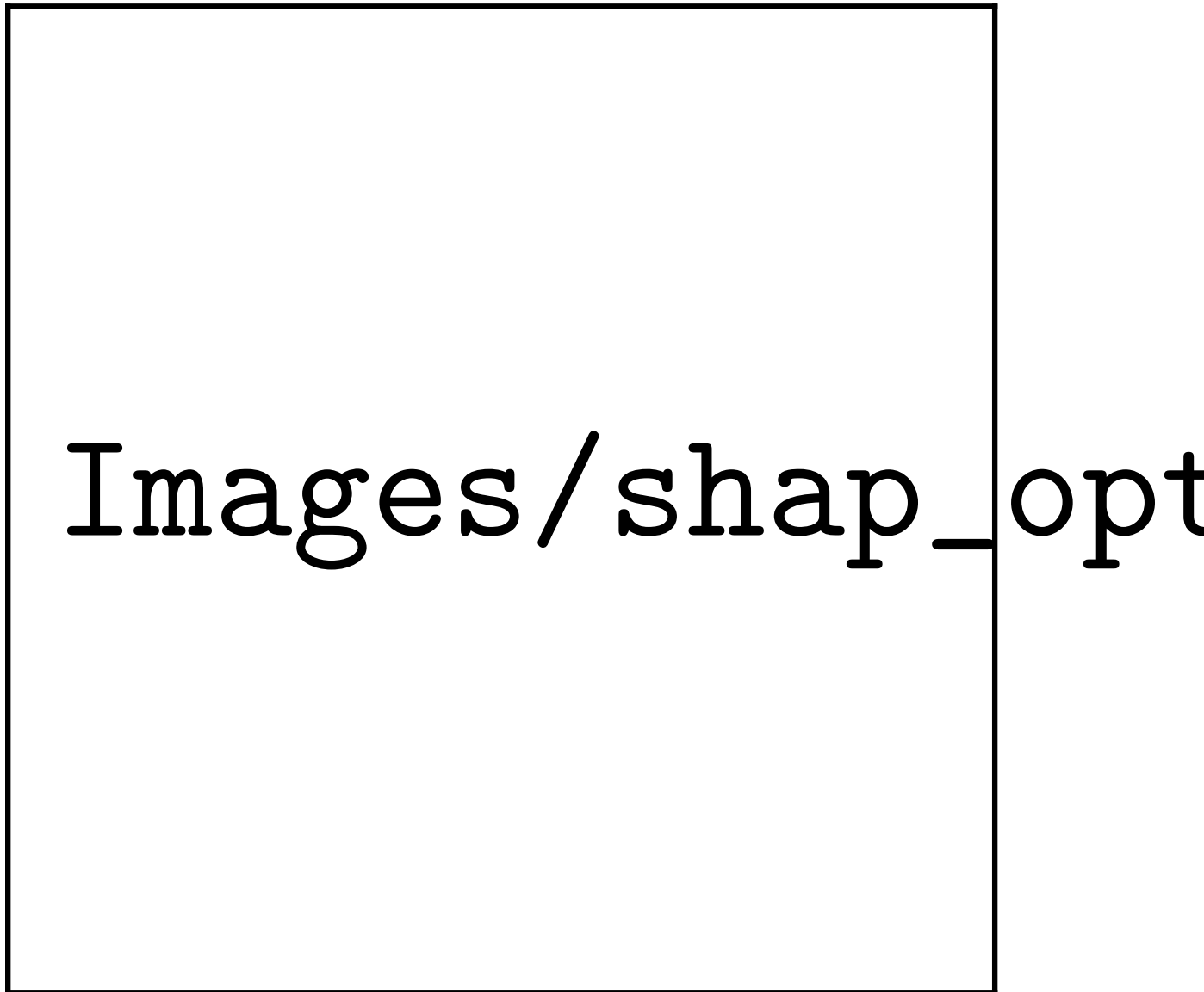


Figure 11: SHAP values distribution by feature (improved model).

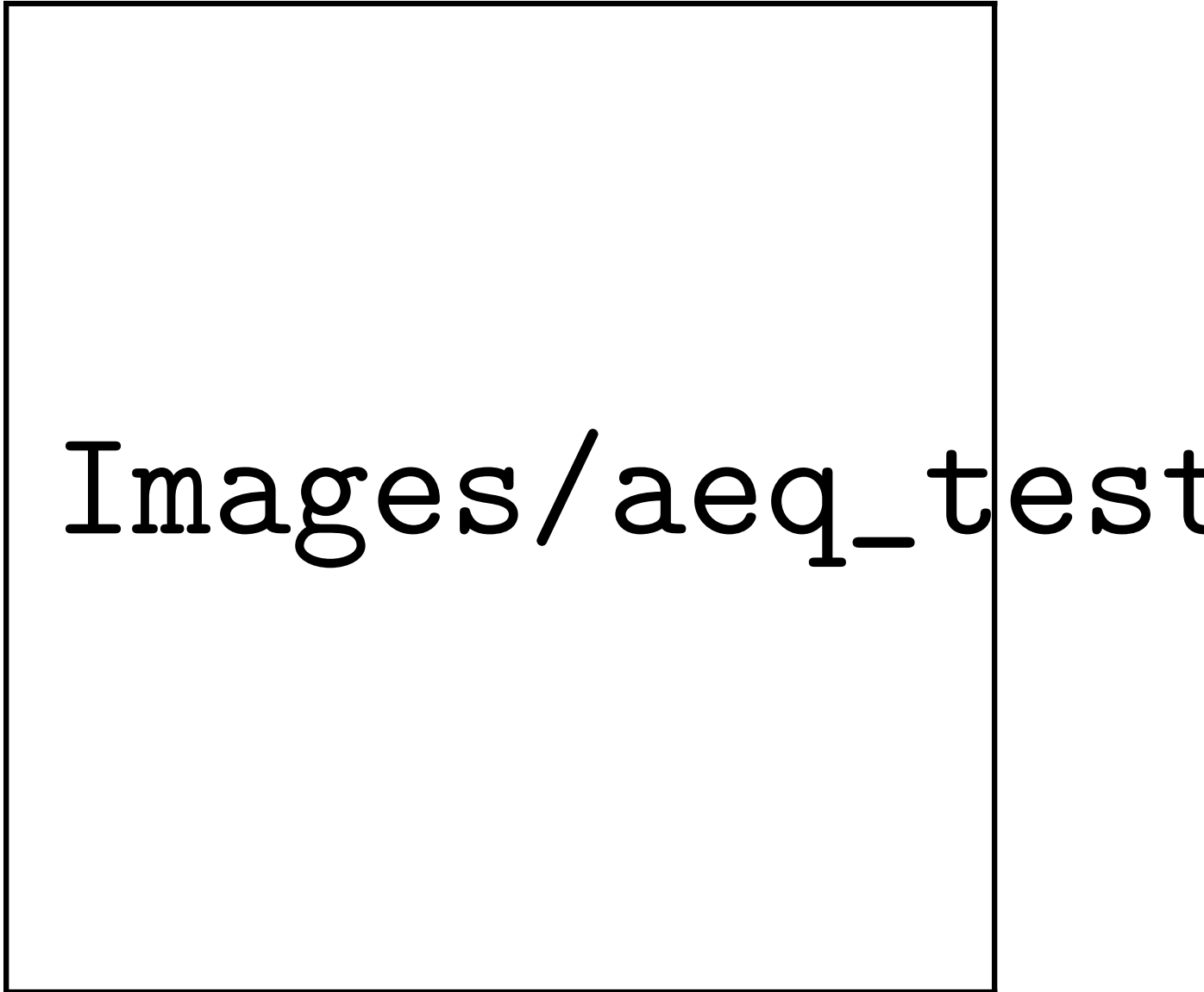


Figure 12: Model assessment for the reference group.

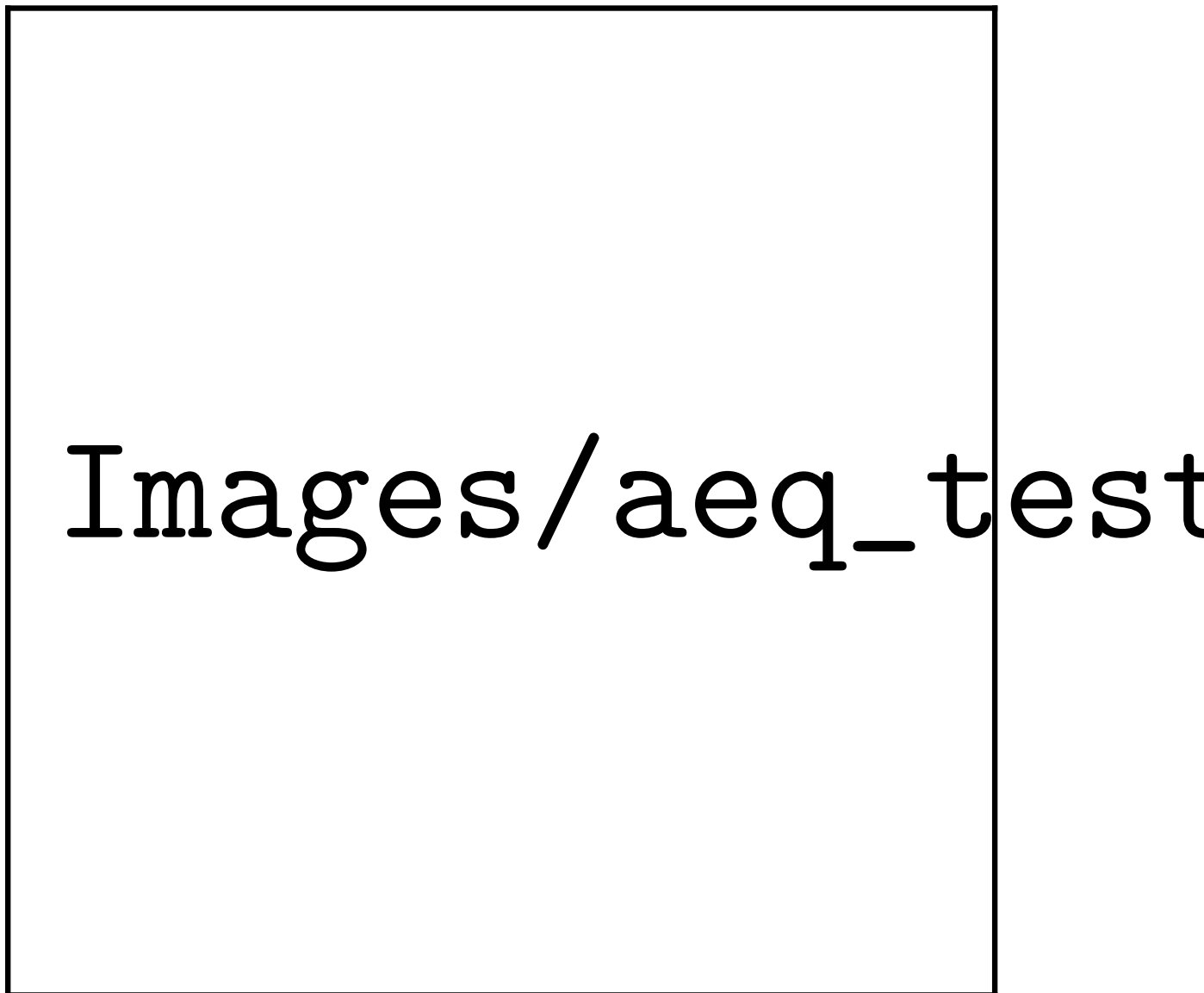


Figure 13: Model assessment for the major group.



Figure 14: Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.1235	0.8765	0.4201	0.3333
Arica y Parinacota	0.0806	0.9194	0.3540	0.2857
Atacama	0.2432	0.7568	0.0625	0.4706
Aysén	0.0000	1.0000	0.2700	1.0000
Bio Bío	0.3642	0.6358	0.1750	0.2903
Coquimbo	0.0890	0.9110	0.2180	0.5854
La Araucanía	0.1256	0.8744	0.2815	0.1905
Libertador Bernardo O'Higgins	0.0841	0.9159	0.1943	0.2800
Los Lagos	0.4921	0.5079	0.4945	0.4312
Los Ríos	0.2642	0.7358	0.1674	0.2632
Magallanes y Antártica Chilena	0.0294	0.9706	0.2538	0.8333
Maule	0.0735	0.9265	0.2190	0.1176
Metropolitana	0.4225	0.5775	0.1952	0.2644
Tarapacá	0.0000	1.0000	0.1728	-
Valparaíso	0.0000	1.0000	0.2411	1.0000
Ñuble	0.0000	1.0000	0.2231	-

Table 16: Bias metrics between attribute values for the base model.

Table ?? and Figures ??, ?? and ?? are the assessment results for the improved model. Table ?? is the crosstab for the variable “Region”.

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.001	1.075	0.999
Major Group	1.004	1.078	0.993
Min. Metrics Group	1.362	4.302	1.619

Table 17: Average disparities for the improved model.



Figure 15: Model assessment for the reference group.

4. Discussion

4.1 Drug Trafficking

The performance of the model shows clear differences between the base and improved configurations. As it can be seen in Table ??, the base model excels in accuracy and precision, but the recall is quite low, indicating a tendency towards negative predictions. The improved version, on the other hand, opts for a compromise, favouring a balanced distribution of false negatives and positives, although at the expense of some accuracy and precision.

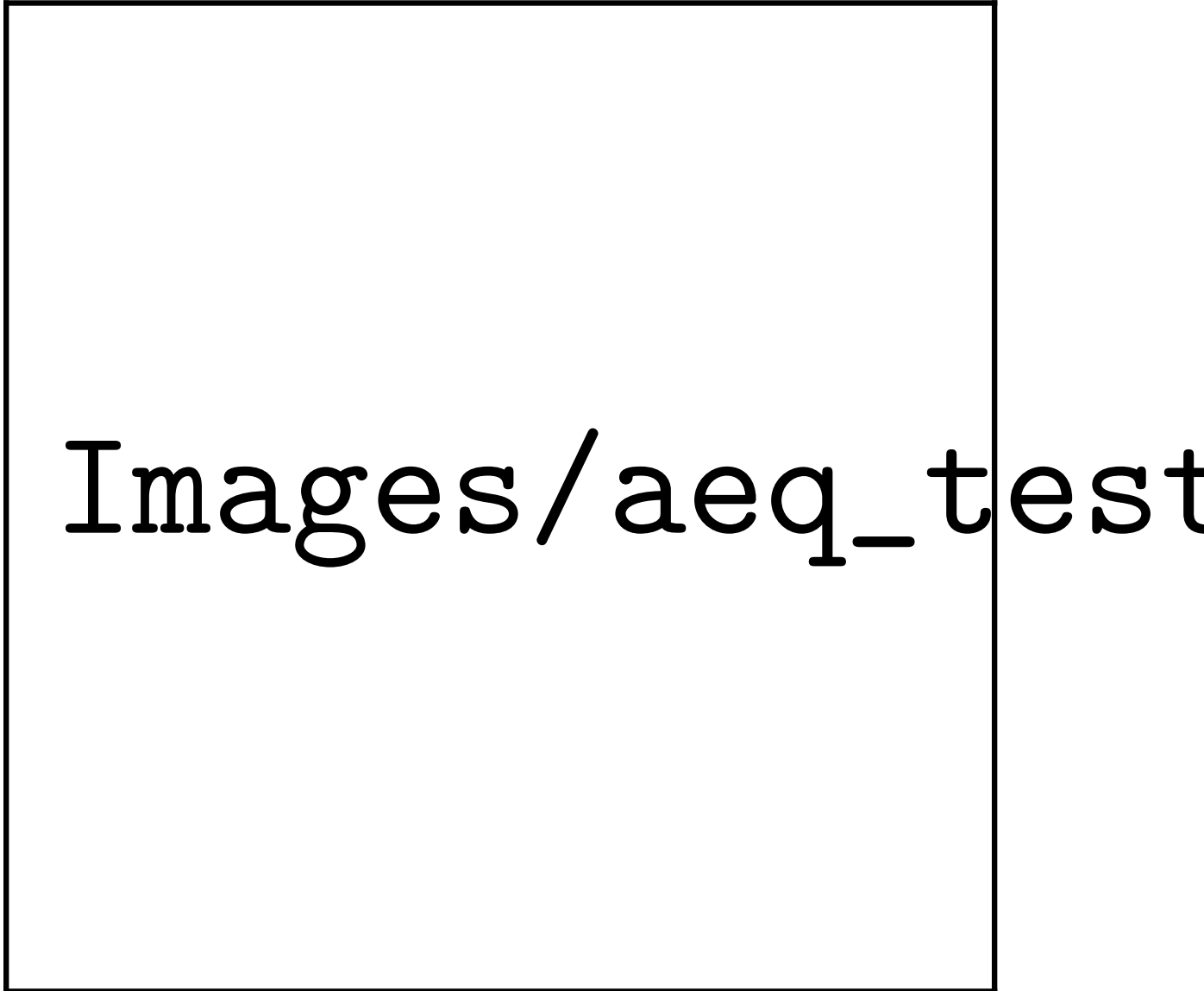


Figure 16: Model assessment for the major group.

Examining the base model SHAP values in Table ??, it becomes evident that the variables “Foreigner” and “Region” predominantly influence the model, with “Barrister” and “Age” trailing them. Particularly, when looking at Table ??, it can be seen that “Foreigner” has on average a negative impact towards predictions, whilst “Region” has a mainly positive one, although given its value, its impact is virtually negligible. “Barrister” and “Age” also have mainly a negligible impact on the predictions. In the improved configuration, the influence of “Region” and “Barrister” equilibrates, but their influence becomes more pronounced than in the baseline setup, as their SHAP values went up from 0.016019 and 0.006392, to 0.046972 and 0.051984, respectively – which is a 193.17% increase for the variable “Region”, and a 713.23% for “Barrister”. On the other hand, “Foreigner” reduces its

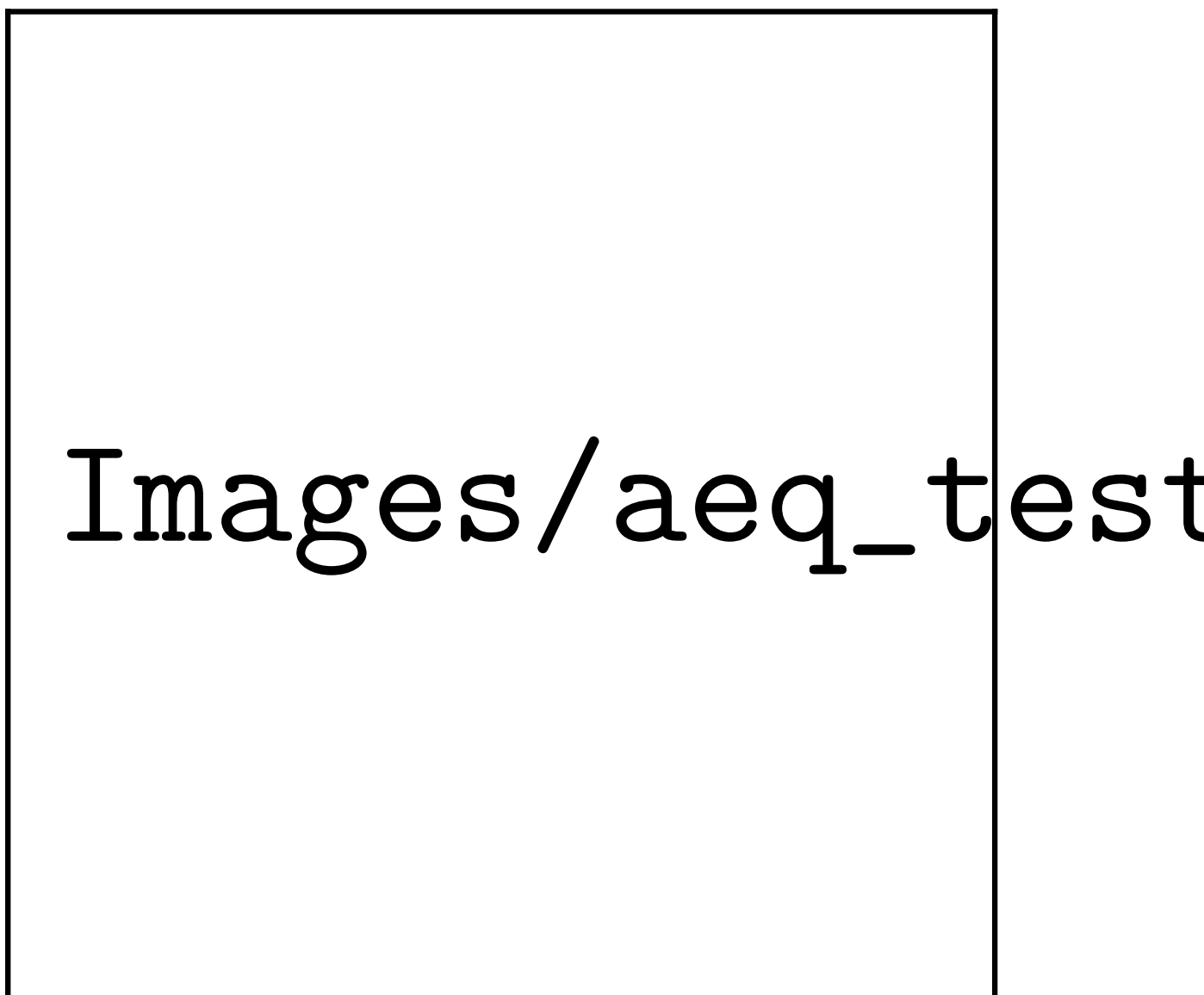


Figure 17: Model assessment for the min metrics group.

impact by 34.15%, and now it has a positive influence towards predictions when looking at Table ??, as its average value is now positive. “Region” still has a net positive impact in the predictions, and its value is now 22.9 times bigger than in the base model.

Analysis of Figures ?? and ?? reveals the dual nature of the “Foreigner” variable, which could positively or negatively influence the model. In the base model, lower value regions leans towards negative predictions, a likely reflection of overrepresentation of certain regions, especially from the northern areas. The improved model, however, moderates this, resulting in a more balanced influence of this variable on predictions. The variable “Barrister” also shifts in influence, moving from a negligible negative impact in the base model to a still

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.4938	0.5062	0.3154	0.2593
Arica y Parinacota	0.2581	0.7419	0.3407	0.5152
Atacama	0.5946	0.4054	0.0380	0.6857
Aysén	0.5556	0.4444	0.2000	0.6512
Bio Bío	0.4768	0.5232	0.1535	0.3543
Coquimbo	0.4136	0.5864	0.1750	0.6030
La Araucanía	0.3571	0.6429	0.2549	0.5167
Libertador Bernardo O'Higgins	0.3318	0.6682	0.1592	0.4779
Los Lagos	0.2292	0.7708	0.4964	0.3245
Los Ríos	0.3585	0.6415	0.1604	0.5250
Magallanes y Antártica Chilena	0.2059	0.7941	0.2411	0.7083
Maule	0.3039	0.6961	0.2243	0.7490
Metropolitana	0.3853	0.6147	0.2043	0.2657
Tarapacá	0.5714	0.4286	0.1091	0.6923
Valparaíso	0.2802	0.7198	0.2040	0.5526
Ñuble	0.3571	0.6429	0.1989	0.7143

Table 18: Average bias metrics between attributes for the improved model.

neutral but positive influence post-optimisation, which goes in hand with what was shown in Table ??.

Insights from Table ?? highlight the base model’s proficiency in positive predictions but its struggle with negative ones. Specifically, its high FNR stands out. The improved version addresses this but experiences an uptick in the FPR — from 0.13% to 39.25% — in contrast to a decrease in FNR, from 40.54% to 31.63%. Table ?? complements this, as the number of true positives went up from 1594 to 1833, while false negatives were reduced from 1087 records to 848 – an approximate 22% reduction; by contrast, the amount of false positives is roughly 304 times bigger than in the base model, as the it rose from 3 to 913 records.

In the base model, as evidenced by Table ?? and corresponding Figures ?? to ??, notable disparities emerge. Its performance metrics indicate consistent failure, with pronounced disparities, especially in the latter metrics. The “Foreigner” variable is a lone exception, achieving FOR parity. Focusing on the critical “Region” variable (Table ??), only a few regions meet the benchmark, but the average FNR per attribute is 74.86%, which means that the model tends to incorrectly predict true positives as negative depending on the region of the defendant in 74% of the cases. The average TPR is 25.13%, which means that the model also has problems in correctly identifying true positives. Finally, the FOR is 34%, meaning that of all the negative predictions made by the model, 34% were actually positive and were erroneously missed. When extending the analysis to the “Zone” variable, disparities in FNR and TPR become even more apparent, underscoring significant regional discrepancies.

Contrastingly, the improved model shows important improvements, as seen in Table ?? and its associated figures (??, ?? and ??). It exhibits a marked reduction in FNR disparities

by an average of 44.37%, FOR disparities by 8.50%, and TPR by 94.67%. Although the model has not attained complete fairness or statistical parity, it represents a significant improvement. The “Region” variable, while still problematic, displays improvement as more regions clear the assessment, as its average FNR was reduced to only 39.76%, while the true positive rate was improved from 25.13% to 60.23%. The average FOR was slightly impacted, as its average value went up to 37.75%. Considering the zones, all now align with the set fairness criteria. In terms of parity, the “Foreigner” variable achieves statistical parity, and “Zone” accomplishes FOR and TPR parity.

4.2 Petty Theft

The results of this experiment, as described in Table ??, show a suboptimal performance for both models, due to the lack of key variables in the dataset that are essential for effective training. However, the improved model shows a clear advantage, outperforming the base model with superior recall and F1 values. While accuracy and precision show marginal fluctuations, the overall performance of the improved model makes it superior.

Insights from Table ?? reveals that the base model predominantly relies on the number of hearings to drive decisions. The regional aspects and the development grade of the crime come secondary, whereas factors like age and barrister contributes the less. Table ?? shows that both “Hearings” and “Barrister” have a negative influence towards predictions, as its average SHAP Value is negative. The improved model, according to Table ??, displays subtle alterations, with the “Development” variable slightly edging over “Region”, as its impact is now 1.12 times bigger than in the base model, while the second variable actually reduced its effect by 2.02%. However, the amount of hearings is still the dominant variable, having a negligible change compared to the base model (a reduction of 0.79%). When looking at Table ??, “Region” now has a more neutral effect, as its SHAP value was reduced by 54.29% (0.023519 to 0.010753), while, on the other hand, “Hearings” is now 1.13 times more influential towards negative predictions. The overall results shown in this table tells us that the optimisations made the variables have a more “negative” effect on predictions, as every feature, excepting “Development”, had a reduction or maintained a negative SHAP value.

Figures ?? and ?? basically confirms what was discussed previously, although it can be seen that “Hearings” had plenty of cases with big positive effect on the predictions, this makes sense as the minimum SHAP value for this variable was -0.455377 for the base model and -0.464773 for the improved one, while the maximum value was 1.438100 and 1.414085, respectively. It can also be seen that an increased number of hearings tends to enhance predictive results. As for the variable “Region”, there is no visible tendency or skew towards a kind of prediction, independently of the region.

Table ?? shows similarities between the situations observed in this and the prior experiment. The base model has problems with false negatives – as its FNR is at 73.20% –, predicting many outcomes as negatives that should have been positives. This manifests in the model’s high recall. The improved model, while attempting to rectify these issues, still struggles with accurate predictions, likely due to the dataset’s absence of crucial features needed for training. In this case, the FNR went down to 64.16%, as Table ?? indicates that the number of false negatives were reduced from 2398 to 2102 records (a 12.34% reduction).

The false positive rate, on the other hand, went up from 4.96% in the base model, to 12.21% in the second one, as the amount of false positives rose from 432 to 1064 predictions (i.e., more than doubled).

A comparative analysis using Table ?? and related Figures ?? to ?? suggests that the metrics of the base model are slightly better than in the previous experiment. Despite this, it still registers failure on most tests. However, “Zone” and “Age” achieve FNR parity, and the former also achieves TPR parity. The protected variable “Region” underperforms in many areas. When looking at Table ??, it can be seen that the FNR between attribute values is also high, as its average is at 85%, meaning that more than 8 out of 10 predictions that were classified as negatives were actually positive, denoting big accuracy problems in between regions. TPR is also low, as it is on average at 14.05%, meaning that the model also has problems when classifying positives. The false omission rate, however, stayed relatively low with an average of 24.64%.

With the improved model, disparities do improve, but the magnitude of the change is not as profound as in the previous experiment. Specifically, according to Table ?? and Figures ?? to ??, FNR disparities were decreased by an average of 25.40%, while FOR disparities decreased by 7.04%. However, TPR disparities actually increased by 11.71%. The model is still unable to attain overall fairness or statistical parity. Nevertheless, every variable, with the exception of “Region”, aligns with FNR parity. “Zone” and “Age” are the sole achievers of TPR parity, with “Age” being the only one that achieved FOR parity. Despite improvements, “Region” does not have any parity alignment, even though its average bias metrics between attributes were also improved overall, as the FNR decreased from 85% to 61.41%, its TPR went up to 38.58%, and its FOR went down to 21.72%, therefore aligning with the statement made previously: the model is not only fairer, but it is also more accurate than the first one.

5. Conclusion & Future work

The improvements made using AI Fairness 360 and Fairlearn were mainly positive, as both experiments improved their overall fairness. Biases and disparities by attribute were reduced, although these changes were not sufficient to achieve statistical parity and/or fairness.

The changes also improved the parity levels between our protected variable: Region, which means that the inequalities between the different regions of the country were reduced, thus minimising potential biases.

Most importantly, these changes did not have an overall negative impact on model performance. In the first experiment we obtained a slight reduction in both accuracy and precision, but in the second experiment the model actually improved in terms of performance.

But this prompts the question: to what extent do the models contribute to addressing these biases and inequalities? The response is likely not much in comparison to the data that is used for training. This is the primary conclusion drawn from these two experiments: the level of fairness depends more on the data itself than on the trained model. Essentially, as the model learns from the data patterns, it is impossible to train something that disregards these issues.

The work done by this and other frameworks is mainly to mitigate these problems, not to avoid them altogether. If something like this is wanted or needed, it can be done in two ways: one is to adapt the way in which data that is already stored in a database is collected. This mainly involves creating more complex queries that already take these ethical aspects into account, although this is usually impractical. On the other hand, it can be beneficial to utilise data cleansing, quality checks, and data augmentation methods to minimise ethical concerns. One such approach would be to implement sampling techniques or generate synthetic data (e.g. SMOTE), although the obvious problem with this is that it can introduce noise into the model.

However, this is just one aspect of the issue. Quantity is crucial, as it is usually needed a dataset with a substantial number of entries for training, as well as having multiple variables that can potentially be utilised for these purposes, including those for doing some feature engineering. In this instance, we lacked features that may have been advantageous for improving the prediction accuracy of the models in both experiments. In particular, the analysis lacks critical variables, including the sex of the accused, real age, previous cases, and other relevant case information such as days, expert reports, and degree of participation.

Ultimately, while some improvement techniques can enhance models, data remains the primary factor in developing a responsible and impartial model. Thus, altering hyperparameters or adjusting prediction thresholds may prove insufficient when the available data is lacking, as has been demonstrated by these experiments.

Finally, as we continue to develop and improve our methodology for identifying and addressing biases in ML models, a number of possibilities emerge for future investigation and enhancement of our current framework.

1. **Optimisation Techniques in Model Training:** There is potential in using regularization methods that take into account protected variables (?). By incorporating fairness considerations into the optimisation process, it is expected that models will be able to be trained to intrinsically decrease bias and more accurately represent all aspects of the data. This would have a twofold benefit, improving prediction precision while also guaranteeing that the model does not unintentionally continue societal biases.
2. **Data Disparity Sampling Techniques:** Data forms the fundamental basis upon which ML models are built. However, discrepancies in data may create a notable source of prejudice, particularly when certain groups are underrepresented. To overcome this hurdle, we could examine and utilise advanced sampling techniques customised to the distinct needs of each use case. By guaranteeing that the training data is a more representative sample of the wider population, we can create models that are both more precise and impartial. These techniques may vary from oversampling underrepresented groups to more advanced methods of creating synthetic data, guaranteeing that models have enough data to learn from even if the historical datasets are biased.

Acknowledgments

The authors would like to thank Chile's Public Criminal Defender's Office for providing access to the data used in this study. This research was supported by IDB Lab, innovation laboratory of the Inter-American Development Bank Group [Project ATN/ME-18240-CH], National Agency for Research and Development, ANID + Applied Research Subdirection/IDeA I+D 2023 grant [folio ID23I10357], ANID PIA/BASAL FB0002, ANID/PIA/ANILLO ACT210096, CH-T1246 : Oportunidades de Mercado para las Empresas de Tecnología - Compras Públicas de Algoritmos Responsables, Éticos y Transparentes.